

Deep Learning, Big Data and High Performance Computing to enhance Business and Marketing, while warying of Fake News: Challenges, Opportunities and Future Works*

*Note: Sub-titles are not captured in Xplore and should not be used

1st Zair Bouzidi

Laboratory LMA,
Commercial Sciences Dept,
Faculty of Economy, Bejaia University
Bejaia, Algeria
zair.bouzidi@univ-bejaia.dz

3rd Abdelmalek Boudries

Laboratory LMA, Commercial
Sciences Dept, Faculty of Economy,
Bejaia University
Bejaia, Algeria
abdelmalek.boudries@univ-bejaia.dz

2nd Mourad Amad

LIMPAF Laboratory, Computer Science
Dept, Faculty of Science and Applied Science,
Bouira University
Bouira, Algeria
mourad.amad@univ-bouira.dz

Abstract—The Gathered or deducted knowledge managed to be applied to multiple domains including notably social networks, forecasting, marketing, business, detecting fraud and financial time series prediction, using deep learning paradigms, constitute our major challenge. Many contents during events convey timely actionable and precious information. First responders, political and economic decision makers, and even the public can use certain information, as awareness, to gain insight into the situation as it unfolds. However, processing information from big data (including smart data and social media) to obtain such information, involves solving multiple challenges, notably parsing informal contents, handling information overload and prioritizing types of relevant information, while being wary of fake news.

Index Terms—Big data, Business, Deep learning, Fake news, Marketing, Supercomputing

I. INTRODUCTION

Since recent years, social networks have become more and more ubiquitous in our daily lives. They have a potential resource to communicate, detect, track and extract information to improve the management of our daily life [1], [2], with the ability to share a message with a potentially large audience. However, the amount of information that is now released, especially during critical times (crisis situations), can make it difficult to locate precious, useful and usable information, while being wary of fake news [3], [4]. During the various crisis events in recent years, social media has enabled the population to quickly release a considerable amount of information about current events to the public: this will also serve to help managers make good and quick decisions [2]. That became standard practice for communicating information in a timely manner [1].

The link between social media and notoriety is the subject of increasing [6] attention from researchers. Many studies

have been carried out in recent years [6] for the automatic identification of different types of content. With the proliferation of social media, an ongoing event is being discussed across all of these channels. To get a complete view of the event, even to avoid certain fake news, it is important to retrieve information from multiple sources that usually contain qualitative differences, and possibly even detect fake news. However, when it comes to retrieving information posted on social networks, the task is daunting.

The Web has several sources of information on which an ongoing event is discussed. Several automated systems have been designed to help managers identify and filter useful information from multiple sources [7]–[9]. Data is often available from different sources, namely databases, log files, online Web applications, and social media networks. Similarly, data, in the area of Bank and tax department, are generated from numerous sources, including laboratory experiments, accounting datasets, bank records, insurance and crisis data which are accessible online.

The rest of the paper is organized as follows. Chapter 2 contains background with the latest surveys and related works. In chapter 3, we begin this survey with raw data and their storage, big data and smart data. Chapter 4 illustrates an overview on supercomputing. A complete study on the different models of neural learning constitutes chapter 5. Fake news constitute content of Chapter 6. Chapter 7 relates all that is improvement of the economic world with all the advantages and very few disadvantages of big data, supercomuting and in particular of deep learning, while warying of false news. We end with a conclusion and many future works.

II. BACKGROUND AND RELATED WORKS

The table 1 shows many surveys for Big Data [10]–[15], and in particular in the field Smart Data and Social Media

[16]–[20]. In Deep Learning, we have [10], [16], [22]–[26]. Even the Parallel Computing Environment has been covered [13], [20], [27]–[31]. Some studies have also touched on the Evolution of FPGA [30], Human Computing and Machine Learning to Make Sense of Big Data [32], Decision Making and emerging Big Data [33], Quality of Social Media Data in Big Data [34] and Big Data privacy in public Social Media [35].

There are also studies that addressed data acquisition for Recent Neural Learning Approaches of Marketing, Business, Finance and Fraud detection, from Twitter only [36]–[38], Twitter & Facebook [7], [8], all the Web [9], [41] and Connected Objects (Smart Data) and All the Web [39], [40], [59].

Other studies have focused on Economic Tasks such as Detect novel frauds [42], Trading performance [43], Exchange rate prediction [44], Stock prediction [45], Trade on the stock market [46], Company stock prices [47], Forecasting of financial time series [48] and Fraud detection and Financial Time Series Prediction [41], [49], [50].

For fake news studies, we have review of Detecting and Fighting Fake News [3]. All of these studies only looked at work done only on Big Data, Deep Learning, and Supercomputing separately.

Only our investigation will cover, at the same time, the various models of Deep Learning, including the Capsule Network model and even the updated model of Spiking Neural Network with all the Big Data, Smart Data up to the models of connected objects to arrive at Supercomputing with all the models of Marketing, Corporate Marketing, Business strategies, Fraud detection and Forecasting of financial time series.

A. Recent Surveys

B. Related Works

The next subsections provide instructions on how to insert figures, tables, and equations in your document. The table 1 shows many surveys for Big Data [10]–[15], and in particular in the field Smart Data and Social Media [10], [16]–[20]. In Deep Learning, we have [16], [21]–[26]. Even the Parallel Computing Environment has been covered [13], [20], [27]–[31]. Some studies have also touched on the Evolution of FPGA [30], Human Computing and Machine Learning to Make Sense of Big Data [32], Decision-Making and emerging Big Data [33], Quality of Social Media Data in Big Data [34] and Big Data privacy in public Social Media [35].

There are also studies that addressed data acquisition for Recent Neural Learning Approaches of Marketing, Business, Finance and Fraud detection, from Twitter only [36]–[38], Twitter & Facebook [7], [8], all the Web [9], [41] and Connected Objects (Smart Data) and All the Web [39], [40], [59]. Other studies have focused on Economic Tasks such as Detect novel frauds, Trading performance, Exchange rate prediction, Stock prediction, Trade on the stock market, Company stock prices, Forecasting of financial time series and Fraud detection and Financial Time Series Prediction.

TABLE I
CLASSIFICATION OF RECENT SURVEYS.

Tasks	Surveys
Deep Learning	
Deep Learning	[16], [21]–[26]
Big Data	
Big Data	[10]–[15]
Smart Data & Social Media	[10], [16]–[20]
Supercomputing	
Parallel Computing Environment	[13], [20], [27]–[31]
Evolution of FPGA	[30]
Human Computing and Machine Learning	[32]
Decision-Making and emerging Big Data	[33]
Quality of Social Media Data in Big Data	[34]
Big Data privacy in public Social Media	[35]
Economic Tasks	
Detect novel frauds	[42]
Trading performance	[43]
Exchange rate prediction	[44]
Stock prediction	[45]
Trade on the stock market	[46]
Company stock prices	[47]
Forecasting of financial time series	[48]
Fraud detection and Financial Time Series Prediction	[41], [49], [50]
Deep Time-Series Clustering	[51]
Fake News	
Detecting and Fighting Fake News	[3], [52]–[57]
Detecting Fake News Spreaders	[58]

All of these studies only looked at work done only on Big Data, Deep Learning, and Supercomputing separately. Only our investigation will cover, at the same time, the various models of Deep Learning, including the Capsule Network model and even the updated model of Spiking Neural Network with all the Big Data, Smart Data up to the models of connected objects to arrive at Supercomputing with all the models of Marketing, Corporate Marketing, Business strategies, Fraud detection and Forecasting of financial time series.

III. FROM RAW DATA TO SMART DATA VIA BIG DATA

Data ingestion is the process of collecting raw data from various silo databases or files and integrating it into a data lake on the data processing platform, e.g., Hadoop data lake. Today, several data ingestion tools are available for ingesting a variety of data onto Hadoop.

Data processing platforms are Data Lake and Data Warehouse where the first one is a storage repository that holds a huge amount of raw data in its native format. It has the schema-on-read characteristic. It is thus more agile as data can be easily configured and reconfigured following different models during the analysis, as Hadoop but the data structure and requirements are not defined until the data is used and typically stores data using a flat architecture. While the second one is a storage repository that holds a huge amount of raw data in its native format.

A. Big Data

Big Data refer to all the digital data produced by the use of new technologies for personal or professional purposes, including [12], [32], [35] corporate data (emails, documents, databases, business processor history, etc.) as well as data from sensors, content published on the Web (images, videos, sounds, texts), e-commerce transactions, exchanges on social networks, data transmitted by connected objects (electronic labels, smart meters, smartphones, etc.), geolocated data, etc.

They include information published on social networks (age and contact details, reviews, comments, photos, videos), Web browsing (through the famous cookies) and online purchases. Massive data duplication is one of the keystones of the big data architecture. They provide very interesting clues about consumer behavior and market trends, making it possible to know your profile, but also your overall behavior (frequency of use of social networks and of your online purchases (history of transactions, expenses made), channels used, hours of connection, etc.). Cloud computing, hybrid supercomputers (high performance computing or HPC) and distributed file systems (DFS) are among the main storage models currently available. Big data is an essential tool for BtoB and BtoC companies. The data collected helps them to design personalized marketing campaigns adapted to the needs, preferences and behavior of consumers. This information helps improve the customer experience, attract prospects and retain existing customers. Improved targeting makes marketing campaigns more effective and reaches the desired segment, the one that is most likely to be interested in the company's products and / or services. Big data is also a competitive advantage for professionals who hold a multitude of data, because they can anticipate changes in behavior and better understand why consumers have turned to a particular service provider. Big data is a valuable tool in vast private and public fields ranging from online sales to scientific research, including culture, politics (electoral campaigns), transport, insurance, industry, the banking sector and energy. Online Listening Tools tracking

TABLE II
COMPARATIVE TABLE OF ALL TECHNIQUES AND METHODS USED IN MODELS.

Ref	Identification Methods	Used Methods
[7], [8]	Based on Artificial Neural Network	Social Media (Twitter & Facebook)
[9]	Based on FeedForward Neural Network	Social Media (All the Web)
[41]	Based on Hybrid of Deep CNN-LSTM (DeepCNN-LSTM)	Social Media (All the Web)
[39]	Based on Recurrent Neural Network	Smart Data & Social Media (Entire Web)
[40], [59]	Based on LSTM	Smart Data & Social Media (Entire Web)

automatically Channels to collect Contents are Contents are collected from all online channels tracked automatically by the Online Listening Tool, namely Radian6 or any its competitor,

such as Awario, Brand24.com, Brandwatch, Mention, Keyhole, Socialert.net, SocialPilot.co, Simplify360, etc. from websites to social media, such as Twitter, Facebook, LinkedIn, Instagram, Google+, Youtube and so on [17], [60]. Actually, many networking platforms allow access to their data via Application Programming Interface (API) [16].

B. Smart Data

Knowledge is thus generated from the data, based on technologies making complex processing and generating knowledge, a source of added value to become the foundation of the new data economy [13], [29]: that's the reason for which big data must be transformed into smart data. It is the information produced every day, in all situations of life (at workplace, doctor, behind the wheel of car or baker, via smartphone, smartwatch or computer). It can be textual data, images or sensors. If this data can be analyzed, processed and used, we can really speak of it as raw material. These masses of data, big data, contain immense potential. Streaming is a way of broadcasting and reading streaming content (widely used on the Internet). It involves being connected to internet server, opposed to the file download requiring to recover all the data of a file.

TABLE III
SMART DATA APPLICATION AREAS.

No	Smart Data application areas	Smart Data application
1	Financial services	Fraud detection and prevention
2	Retail	Allowing brands to: - Analyzing the sentiment of their customers, and - Offering personalized and contextual promotions
3	Telecommunications industry	Possibility of - Better allocating bandwidth based on real-time needs, and - diagnosing the condition of antennas
4	Manufacturing	preventive maintenance
5	Healthcare	- Monitoring patient vital signs and - Reducing readmission rates
6	Oil industry	- Proactive repairing infrastructures, and - Balancing the power delivered according to consumption
7	Public sector	- Detecting and preventing intrusion attempts on the network, and - Predicting the risk of epidemics
8	Transport sector	Possibility of detecting risky conduits

Smart data is based primarily on real-time data analysis. It refers to an approach to data analysis analyzing directly data at the source, as a different concept of Big Data and without the need to transmit it to a centralized system. Big and smart data are transversal technologies: will impose themselves in all areas of life by, changing them for life. Data are collected, converted, placed in databases and processed in waves, but usually out of date once analyzed. They refer to the analysis of data directly at the source to enable immediate decisions, without sending them to a centralized system. Processing time is reduced to just seconds. While Smart referring to the intelligence, thanks to processing less data (statistical models are

responsible for determining the relevant variables, even most correlated). It enables to control and qualify data: used for establishing a coherent marketing strategy. Generating leads to increase sales in building customer loyalty by providing optimal experience and satisfaction, unlike Big Data.

The possibilities offered by Smart Data are therefore numerous. Table 3 shows some among Smart Data application areas. Virtually all areas of society and the economy can benefit from the intelligent use of data. Big data technologies are already widely used in finance and marketing. Data scientists, marketers, and manufacturers are experimenting with ways to use edge computing and smart data devices to generate more revenue, improve decision-making, and spot equipment issues before they go. Smart Data can also be useful in many industries.

IV. SUPERCOMPUTING

Supercomputing is now as a continuum of computing. There is definitely a big need for high capacity systems to enable applications that need precision and capacity. For these high-capacity models to work it needs composable systems that can permit to compose a solution by automating workflows, using all models of supercomputing. That means supercomputing is becoming even more useful to research, among others. The term supercomputing refers to the processing of massively complex or data-laden problems using the concentrated compute resources of multiple computer systems working in parallel (i.e. a supercomputer). Supercomputing involves a system working at the maximum potential performance of any computer. Sample use cases include weather, energy, life sciences and manufacturing, health care, social networks, mining, recommendation systems, image processing, pattern recognition and all the predictions.

An information technology forecasts that supercomputing will have a significant impact on the data processing operations of corporations and be a competitive necessity actually [31]. In current supercomputers, storage is typically provided by parallel distributed file systems for hot data and tape archives for cold data.

TABLE IV
COMPARATIVE TABLE OF ALL MODELS USING SUPERCOMPUTING.

Models	Identification Usage of Supercomputing
[13], [20], [27]–[31]	Parallel Computing Environment
[30]	Evolution of FPGA : FPGAs inherent reconfigurable nature & low power consumption
[32]	Human Computing and Machine Learning to Make Sense of Big Data
[33]	Decision-Making and emerging Big Data
[34]	Quality of Social Media Data in Big Data
[35]	Big Data privacy in public Social Media

These file systems are often compatible with local file systems due to their use of the POSIX interface and semantics, easing development and debugging because of easily running both on workstations and supercomputers. There is a wide

variety of file systems, each tuned for different use cases and implementing different optimizations. However, the overall application performance is often held back by I/O bottlenecks due to insufficient performance of file systems or I/O libraries for highly parallel workloads. Performance problems are dealt with using novel storage hardware technologies as well as alternative I/O semantics and interfaces. These approaches have to be integrated into the storage stack seamlessly to make them convenient to use.

Upcoming storage systems abandon the traditional POSIX interface and semantics in favor of alternative concepts such as object and key-value storage; moreover, they rely on new technologies such as Non-Volatile Memory (NVM) emulation and burst buffers to enhance performance. Additional tiers of storage will increase the importance of hierarchical management. These changes will be disruptive and require developers to modify their approaches to data management and I/O. A thorough understanding of today's storage infrastructures, is crucially important for designing and implementing scalable storage systems suitable for needs of exascale computing [28]. This work integrates the core ideas of deep learning and its applications in application domains of Marketing corporate, Business Strategies, Fraud detection and Financial time series prediction, to be accessible to all scholars and researchers [28]. Field Programmable Gate Arrays (FPGAs) inherent reconfigurable nature and their low power consumption have become so complementary to microprocessors: they are included in few mainstream computer systems for accelerating application specific performance [27], [30].

Deep Learning and Parallel Computing Environment for Bioengineering Systems delivers a significant forum for the technical advancement of deep learning in parallel computing environment across bio-engineering diversified domains and its applications.

V. DEEP LEARNING

In recent years, Deep learning methodologies have achieved impressive results in computer vision [61], speech recognition, image processing, and handwritten recognition of characters. They are beginning to interest the economic world with some very important applications whether in marketing, commerce, fraud and especially finance. While they are currently in their infancy in fault diagnosis [26].

It can also overcome limitations of shallow networks that prevent efficient training and abstractions of hierarchical representations of multi-dimensional training data [26] in many instances. Deep learning offers a set of units such as [62] convolution unit, [63] recurrent unit, and long-short term memory unit for feature extraction on samples with distinct features, delving into the math behind training algorithms used in recent deep networks, explaining existing shortcomings, [26] improvements and implementations.

A. Discussions about DeepLearning Models

Neural learning is carried out by Feedforward or Feedback neural network. In Feedforward, we have supervised learning

TABLE V
CLASSIFICATION OF SOCIAL MEDIA-BASED RECENT NEURAL LEARNING
APPROCHES OF MARKETING, BUSINESS, FINANCE AND FRAUD
DETECTION.

Social Media-based Approches	References
Twitter	[36]–[38]
Twitter & Facebook	[7], [8]
All the Web	[9], [19]
Connected Objects (Smart Data) and All the Web	[9], [19], [39]–[41]

such as Feedforward neural network itself for classification [9], convolutional neural network [36], [37] for image recognition/ classification or Residual neural network (ResNets) [9] for image recognition, and unsupervised learning such as Autoencoder [63] for Dimensionality reduction and encoding, Generative Adversarial Network [38] network for generating realistic fake data, reconstruction of 3D models or image improvement and with supervised or unsupervised learning such as Restricted Boltzmann machine [63] for dimensionality reduction, feature learning, topic modeling, classification, collaborative filtering or many body quantum mechanics. In Feedback, we have only supervised leaning such as Recurrent neural network [39], [40], [59], [63] for sequences recognition as precise timing, Bidirectional Recurrent Neural Network [38] for natural language processing (NLP), Long Short-Term Memory [63] for temporal data such as stock market values over a period of time and video frames, Fully Connected-LSTM [65] for learning non-linear and complex processes in hydrological or meteorological modeling and Bi-Directional-LSTM [37] through time-natural language processing and language translation. Neural learning can be trained in either supervised/unsupervised ways by Radial Basic Function Network [66] for M-means clustering, Least square function, function approximation and time series prediction or unsupervised ways by Kohonen Self Organizing Network [66] for dimensionality reduction, optimization problems or clustering analysis. Figure 1 shows the classification of Deep Learning Models, with their features and limitations: Radial Basis Function Neural Network is supervised and discriminative model but its classification is slow.

While feedforward neural networks are not capable of extrapolation [67], ResNets have increased architectural complexity as a drawback. They are highly dependent on implementing batch normalization layers, as well as adding jump level connections where dimensionality is essential between the different layers. As a CNN variant designed for computer vision image classification tasks, ResNets is proposed to overcome the problems of VGG-style CNNs: stacking convolutional layers (CNNs) to make the model deeper does not guarantee increased the precision of the validation. The ResNet architecture was introduced to ignore connections: by increasing the number of convolutional layers, the generalization and accuracy of validation decreases. CNN does not encode object position and orientation and lacks ability to be spatially invariant to input data: capsule neural networks

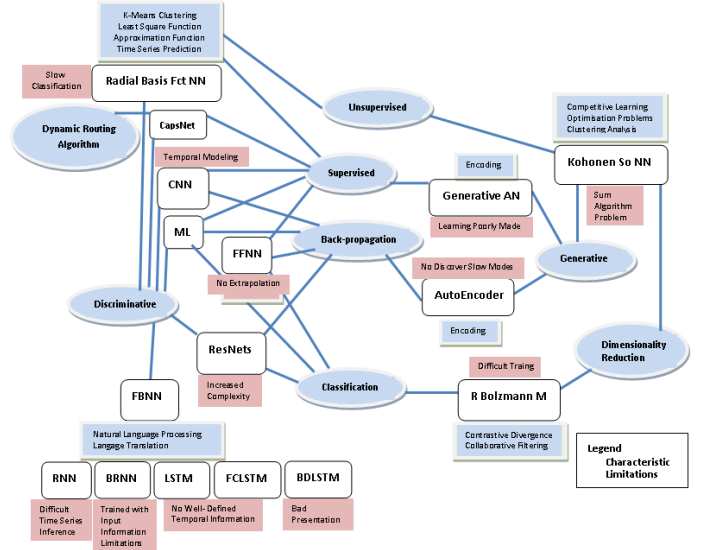


Fig. 1. Classification of Deep Learning with Features and Limitations (Our figure).

(CapsNet) are introduced to solve this CNN problem with the dynamic routing algorithm [68]. Autoencoders are based on deep learning regression for the discovery of slow modes in dynamical systems. However, a rigorous analysis of nonlinear autoencoders is still lacking: they cannot properly discover slow [69] modes. Spiked neural networks are computationally more powerful than other neural network models, in terms of the number of neurons needed [70]. Abiodun et al. (2018) [21] recommend that future research can focus on combining various deep learning models in a single network-scale application, depending on our needs and the characteristics of the different models of deep learning.

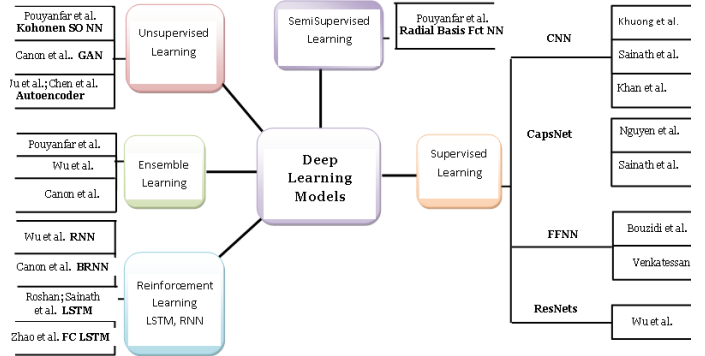


Fig. 2. Classification of Deep Learning Models (Our figure).

Figure 2 shows another form of classification of deep learning models, according to supervised, unsupervised learning, reinforcement and ensemble learning, with the name of the originators of these concepts.

VI. FAKE NEWS

The concept of fake news is a misleading content containing conspiracy theories, fabricated news, satire, rumors and click-

bait [57] (See Figure 3). The dissemination of fake news has become a global issue needing to be attended immediately [71].

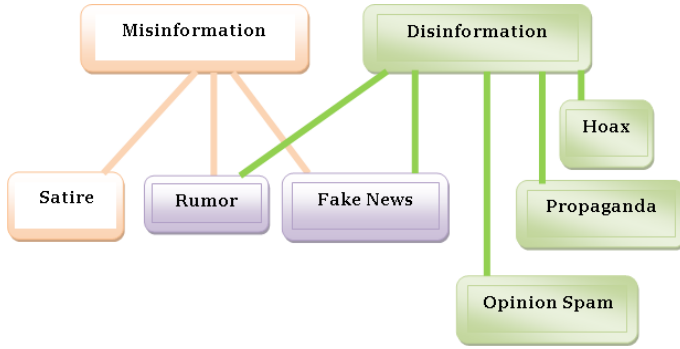


Fig. 3. Fake News Components (Our figure).

Figure 3 shows the different constituents of fake news, namely conspiracy theories, fabricated news, satire, rumors and clickbaits.

False news concept drew major attention during the 2016 US elections [72]. Fake news is both misinformation and disinformation both: false and forged information is disseminated to mislead people or to fulfill a propaganda. It is a vehicle of purposely targeted fabricated news spread to affect the cognitive activities of a user through user-content interaction by indirectly affecting his unconscious behavior. This unconscious behavior, can further strengthen confirmation bias among users and aid to further spreading false news. Humans have always been attracted to controversies and sensationalism [73]: false information about COVID 19 vaccines (dangerous scientific treatment methods) [74], political smear campaigns during elections, etc. In order to clearly understand the dissemination of false news, it is important to examining components that can be divided into four main categories namely social context, content, creator/spreader and target victims [75] (See Figure 4).

A. Types of Fake News

False news may be disinformation, satire, multi-modal, stance, deep fake. Disinformation is misleading and false information to deceive people (political agendas or creating social havoc). Satire (humor and mockery) [78] and Multi-modal (spread of fake news) [79] using multiple means (videos, images, audio, text etc.). Stance can be classified into four types, (agree, disagree, discuss and unrelated) [77]: stance (headline of fake news), whereas a disagreeing stance (contradictory information), Deep fake [80][80] is spread through manipulated video clips, images and recordings (generated by using deep learning techniques). Figure 4 shows an overview of false news, starting by Social Context, Content, Target Victims and creator/Spreader. Social context may be platform (social media or main streaming) or distribution (Community of users or Broadcast pattern). New Content may be Physical (headline, body text or image/video) or Non-physical (sentiment, rumor or new topic). Target Victims may

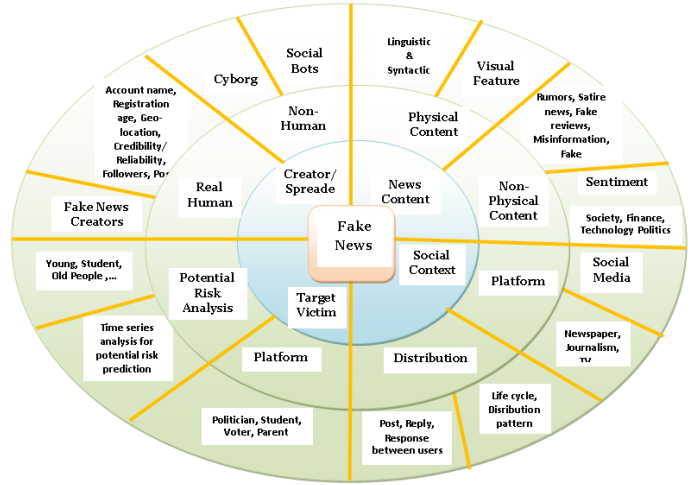


Fig. 4. Fake News Characteristics (Our figure).

be online or main streaming users, young, students, temporal-based analysis. Creator/Spreader may be a human (as Design Author or Publisher) or Non-human (Bot or Cyborg).

B. Features used in Detecting Fake News

Uncovering and recognizing fake news and developing automatic false news detection systems are a challenge: newer methods are usually came up as new solutions [52] to mislead people. The key features in detecting false news are those based on user, temporal analysis, sentiment features, linguistic analysis, social context, and network.

- User-based features include users' profiles that can be analyzed to find out if the person is a false news spreader. They are divided into profile analysis (based on username, age, profile picture, Geo-location and account verification status [81]) and credibility analysis (number of friends and followers): bot accounts have more followers but follow less of them [53].

- Temporal analysis (timing and frequency of posts, user engagement) helps to identify bot accounts [54]: Bot accounts have a more active engagement.

- Sentiment features (triggering emotional response): Bot accounts use out-of context misleading facts (highly polarized and exaggerated) to provoke emotions [3], [4].

- Linguistic analysis involves determining the writing patterns and formats [82]: the excessive use of bold letters in headings and paragraphs, presence of suspicious tokens such as URLs, tags, and excessive uppercase words.

- Knowledge-based analysis involves verifying suspicious information from credible resources such as using verified websites (manually or by using AI algorithms).

- Social context-based analysis includes user network analysis (engagement patterns between online accounts) and distribution network analyses (distribution of information) [55].

- Network features have two types of networks analyzed: Homogeneous and heterogeneous networks. Homogeneous include stance networks (users' stance on a specific idea or news,

with classifying as agreement or disagreement between the main headline and body of the news) and propagation networks [56] (relationship between posts and re-posts): fake news gets re-posted excessively. Heterogeneous involves analyzing relationships between multiple nodes (articles, publishers, users and posts) [57].

C. Fake News Dissemination Studies based on Spreaders Accounts Features

Fake News disseminations are classified based on three categories, namely source, propagation and target based features, as in Figure 5. Source-Based Account Detection: A source of false news can be a human, bot or cyborg. Features of their source are classified into three main categories, namely: feature of personality, historical and of credibility.

Propagation-Based Accounts Detection: A propagator, disseminating false news widely to increase its reach to maximum victims [60], has features classified into three main categories, namely user engagement, time dynamics and platform-based features [83].

Target-Based Accounts Detection: The target features identify end users that are affected by the fake news. A target can be a human, bot or cyborg depending on the nature and domain of false news. Although fake news can reach almost all the users through social media, an easy target will be those people that are more vulnerable and prone to get influenced by the fake news [76]. Victim dynamics mean thoroughly understanding the details of the end user. The details can include age, gender, education history, account creation history, network of followers, location etc. Generally new users with limited exposure to social media are targets of fake news spreaders, as they tend to believe anything presented to them due to lack of exposure [84], as well as Teenagers and aged people with limited knowledge of fake news possibilities on social media are an easy target. Similarly, people with low qualifications and coming from rural areas are more prone to be the victims of fake news [85].

Figure 5 shows the classification of Fake News disseminations, starting features based on by propagation, source and target.

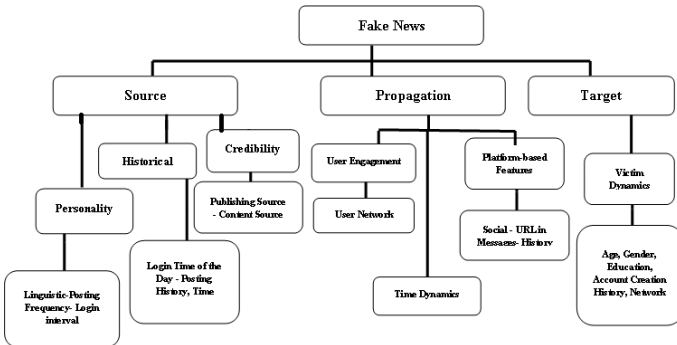


Fig. 5. Spreaders Accounts Features-based Fake News Dissemination (Our figure).

VII. ECONOMIC TASKS: MARKETING, BUSINESS, FRAUD AND FINANCIAL TIME SERIES PREDICTION

With a wide variety of products and buying behaviors, the shelves, on which products are presented, are the most important resources in the retail environment. Retailers can not only increase profits, but also reduce costs by properly managing shelf allocation and product display [86]. Using learning models in the organization of shelves in supermarkets by grouping products that are usually bought together; we can extract the following relation: customers who buy the product X at the end of the week, during the summer, generally also buy the product Y.

There is also the credit institution that permits to decide whether or not to grant credit based on the applicant profile, his / her request, and past loan experiences. The overbooking (optimization of the number of seats in planes, hotels, etc.), the targeting of offers (organization of advertising campaigns, promotions) and the analysis of business practices, strategies and their impact on sales.

TABLE VI
COMPARATIVE TABLE OF ALL ECONOMIC TASKS USED.

Economic Tasks	References
Detect novel frauds	[42]
Trading performance	[43]
Exchange rate prediction	[44]
Stock prediction	[45]
Trade on the stock market	[46]
Company stock prices	[47]
Forecasting of financial time series	[48]
Fraud detection and Financial Time Series Prediction	[41], [49], [50]
Deep Time-Series Clustering	[51]

This knowledge, unknown at first, may be correlations, patterns, or general trends in that data. Table 1 shows a comparison of works with various economic tasks. Experimental data are necessary to verify the correction of the system or the estimation of some difficult parameters to mathematical modeling. Our purpose is to discover, in large amounts of data, valuable information that can help understand the data or predict the behavior of future data. We can use several tools for statistics and artificial intelligence to achieve its objectives: an essential component of Big Data technologies, large data analysis techniques and recently data smart streaming. It is often defined as the process of discovering new knowledge. By examining large amounts of data (stored in warehouses or in streaming) using pattern recognition technologies as well as statistical and mathematical techniques used for the study, we describe the method used for the data collections.

There are also other specific tools, such as the model where some sequence exploration problems lend themselves to the discovery of frequent item sets and their order, traditionally, used in marketing applications to detect patterns among competing elements in large transactions, for example, by analyzing customer shopping basket transactions in a supermarket, according to Han et al. [87]. The models can have many real

applications because the data is encoded as sequences in many areas such as bio-informatics such as genomics and Proteomics [88]. We also see the development of market basket analysis which consists of studying sales (Sales receipt analysis) [23].

Thanks to the huge amount of information available online, the World Wide Web is a fertile ground for research. Its research is at the crossroads of research conducted by several research communities, such as databases, information retrieval and within Artificial Intelligence (AI), particularly the sub-domains of learning and natural language processing [22]. We can use the processing of text corpora to analyze the content and extract knowledge. The main tasks to be accomplished are the recognition of the information presented in the document and its interpretation.

VIII. CONCLUSION AND FUTURE WORKS

This paper exemplifies the comprehensive study of data, of all data: starting from raw to Smart data to Big Data, while emphasizing all economic tasks. This study is supplemented by a overview of supercomputing with a close look at Deep Learning, its evolution and much more about the future with Deep Learning.

In this review, we have not forgotten the advantages of social networks, especially with their popularization in all daily tasks of all populations, but also their disadvantage with the weight of fake news they convey. The truth will be more difficult to decipher, recognize, read, etc., such will be our research perspectives and our future work.

ACKNOWLEDGMENT

We acknowledge support of "Direction Générale de la Recherche Scientifique et du Développement Technologique "DGRSDT"". MESRS, Algeria.

REFERENCES

- [1] Olteanu, A., Vieweg, S., & Castillo, C. (2015), What to Expect When the Unexpected Happens, Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing - CSCW '15. doi:10.1145/2675133.2675242
- [2] Bouzidi Zair (2022), Social networks for Disaster management, Classical Doctorate Thesis, University of Bouira, Algeria, February 06th, 2022
- [3] Bouzidi Z., Boudries A. & Amad M., (2022b), Enhancing Detecting and Fighting Fake News with Big Data and Deep Learning: Challenges, Opportunities and Future Works, 6th International Conference on Information System and Data Mining, ICISDM 2022, Silicon Valley, CA, United States, May 27-29, 2022, www.icisdsm.org, Accepted
- [4] Bouzidi, Z., Boudries, A. and Amad, M., (2022c), Big Data, Deep Learning and Supercomputing to Improve Financial Time Series Forecasting, Fraud Detection, Business and Marketing: Survey, First edition of the International Colloquium on Marketing Management COLIMM22, Algiers, Algeria, Accepted
- [5] Imran, M., Castillo, C., Lucas, J., Meier, P., & Rogstadus, J. (2014b) Coordinating human and machine intelligence to classify microblog communications in crises. Proceedings of (ISCRAM) The 11th International Conference on Information Systems for Crisis Response and Management, (ISCRAM 2014), pp. 712-721, The Pennsylvania State University, Pennsylvania, USA
- [6] Vulic, I. and Moens, M.-F., (2015), "Monolingual and Cross-Lingual Information Retrieval Models Based on (Bilingual) Word Embeddings", SIGIR'15 Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval, Pages 363-372, Santiago, Chile — August 09 - 13, 2015
- [7] Bouzidi, Z., Boudries A. & Amad M. : A New Efficient Alert Model for Disaster !! Management, in Proceedings of Conference AIAP'2018 : Artificial Intelligence and Its Applications, El-Oued, Algeria
- [8] Bouzidi, Z., Amad, M. & Boudries, A. : Intelligent and Real-time Alert Model for Disaster Management based on Information retrieval from Multiple Sources", International Journal of Advanced Media and Communication. 7(4), 309-330, doi:10.1145/253260.253325 (2019)
- [9] Bouzidi, Z., Boudries, A. & Amad, M.: Towards a Smart Interface-based Automated Learning Environment Through Social Media for Disaster Management and Smart Disaster Education. Advances in Intelligent Systems and Computing. SAI 2020. 1228. Springer, Cham, 443-468. https://doi.org/10.1007/978-3-030-52249-0_31 (2020)
- [10] Saroj, Anita & Pal, Sukomal, (2020), Use of social media in crisis management : A survey, International Journal of Disaster Risk Reduction, doi:10.1016/j.ijdr.2020.101584
- [11] Emmert-Streib, F., Yang, Z., Feng, H., Tripathi, S. and Matthias, D., (2020), An Introductory Review of Deep Learning for Prediction Models With Big Data, doi:10.3389/frai.2020.00004
- [12] Saranya, M. and Prema, A., (2017), Survey On Big Data Analytcs using Hadoop ETL, International Journal of Computer Trends and Technology (IJCTT), vol. 48, No. 1, doi:10.14445/22312803/IJCTT-V48P105
- [13] Alexandrov, A., Bergmann, R., Ewen, S., Freytag, J.-C., Hueske, F., Heise, A., Kao, O., Leich, M., Lesser, U., Markl, V., Naumann, F., Peters, M., Rheinländer, A., Sax, M. J., Schelter, S., Hoyer, M., Tzoumas, K. and Warneke, D., (2014), The stratosphere platform for big data analytics, The VLDB Journal, vol. 23, No. 6, pp. 939-964, Springer Berlin Heidelberg
- [14] Sanjeet Singh, Jagmeet Bawa, Gagan Deep Sharma, (2018), The Goal of Business-A Review Paper, Global Journal of Management and Business Research, Volume XV, Issue XI, Version I, 2015
- [15] Mininath R. Bendre & Vijaya R. Thool, (2016), Analytics, challenges and applications in big data environment: a survey, Pages 206-239, Journal of Management Analytics, Volume 3, 2016 - Issue 3, https://doi.org/10.1080/23270012.2016.1186578
- [16] Imran, M., Offi, F., Caragea, D. & Torralba, A.: Using AI and Social Media Multimodal Content for Disaster Response and Management: Opportunities, Challenges, and Future Directions. Information Processing & Management, 57(5), 1-9. http://sci-hub.tw/10.1016/j.ipm.2020.102261 (2020)
- [17] Ruggiero, A. & Vos, M., (2014), Social Media Monitoring for Crisis Communication: Process, Methods and Trends in the Scientific Literature, In Online Journal of Communication and Media Technologies, Vol 4, No. 1
- [18] Poddar, S., Mondal, M. & Ghosh, S., (2020), A Survey on Disaster: Understanding the After-effects of Super-cyclone Amphan and Helping Hand of Social Media, Computer Science, Computers and Society
- [19] Bouzidi, Zair and Boudries, Abdelmalek and Amad, Mourad, (2021), Deep Learning and Social Media for Managing Disaster : Survey, Intelligent Systems Conference 2-3 September, 2021, (IntelliSys 2021), Amsterdam, The Netherlands, Volume 294, Intelligent Systems and Applications, IntelliSys 2021, Vol 1, LNNS 294, Chapter 2, DOI : 10.1007/978 - 3 - 030 - 82193 - 7_2, Springer Nature
- [20] Bouzidi, Z., Boudries, A. & Amad, M., (2021c), Enhancing Crisis Management because of Deep Learning, Big Data and Parallel Computing Environment : Survey, ID:443, Proceedings of the 3rd International Conference on Electrical, Communication and Computer Engineering (ICECCE), 12-13 June 2021, Kuala Lumpur, Malaysia, doi:10.1109/ICECCE52056.2021.9514189
- [21] Abiodun, O. I., Jantan, A., Omolara, A. E., Dada, K. V., Mohamed, N. A. & Arshad, H. (2018). State-of-the-art in artificial neural network applications : A survey. In Heliyon, 4(11), DOI:10.1016/j.heliyon.2018.e00938
- [22] Kosala, R. and Blockeel, H., (2000), Web Mining Research : A Survey, ACM SIGKDD Explorations Newsletter, vol. 2, No. 1, ACM
- [23] Masseglia, F., Teisseire M. & Poncelet, P. (2005). Sequential pattern mining: A survey on issues and approaches. Encyclopedia of Data Warehousing and Mining
- [24] LeCun, Y., Bengio Y. and Hinton G., (2015), Deep Learning: Review, Nature, Vol 521, May 28st
- [25] Nash, W., Drummond, T. and Biribilis, N., (2018), A review of deep learning in the study of materials degradation, Materials Degradation, doi:10.1038/s41529-018-0058-x
- [26] Shrestha, A. and Mahmood, A., (2019), Review of Deep Learning Algorithms and Architectures, IEEE Access, vol. 7, pp. 53040-53065, doi:10.1109/ACCESS.2019.2912200

- [27] Sangaiah, A. K., (2019), Deep Learning and Parallel Computing Environment for Bioengineering Systems, ISBN 9780128167182, publisher Academic Press
- [28] Luttgau, J., Kuhn, M., Duwe, K., Alforov, Y., Betke, E., Kunkel, J. and Ludwig, T., (2018), SURVEY OF STORAGE SYSTEMS FOR HIGH-PERFORMANCE COMPUTING, SUPERCOMPUTING FRONTIERS AND INNOVATIONS, vol. 5, No. 1, doi:10.14529/jsfi180103
- [29] Stillger, M., Lohman, G. M., Markl, V. and Kandil, M., (2001), LEO-DB2's learning optimizer, Very Large DataBases (VLDB), vol. 1, pp. 19-28
- [30] Awad, M., (2009), FPGA supercomputing platforms: A survey, FPL 09: 19th International Conference on Field Programmable Logic and Applications, pp. 564-568, doi:10.1109/FPL.2009.5272406
- [31] Amini, M. M. and Schooley, R., (1993), Supercomputing in corporate America: A sample survey, Information & Management, vol. 24, No. 6, pp. 291-303, doi:10.1016/0378-7206(93)90027
- [32] Offi, F. and Meier, P. and Imran, M. and Castillo, C. and Tuia, D. and Rey, N. and Briant, J. and Millet, P. and Reinhard, F. and Parkan, M. and Joost, S., (2016), Combining Human Computing and Machine Learning to Make Sense of Big (Aerial) Data for Disaster Response, Big Data, vol. 4, No. 1
- [33] Horita, Flavio E.A., Albuquerque, Joao Porto (de), Marchezini, Victor and Mendiondo, Eduardo M., (2017), Bridging the gap between decision-making and emerging big data sources: An application of a model-based framework to disaster management in Brazil, Decision Support Systems, vol. 97, pp. 2-22, doi:10.1016/j.dss.2017.03.001
- [34] Immonen, A. and Paakkonen, P. and Ovaska, E., (2015), Evaluating the Quality of Social Media Data in Big Data Architecture, IEEE Access, vol. 3, pp. 2028-2043, doi:10.1109/ACCESS.2015.2490723
- [35] Smith, M., Henne, B., Szongott, C. and Voigt, G. von, (2012), Big data privacy issues in public social media, 6th IEEE International Conference on Digital Ecosystems Technologies, pp. 1-6, Doi:10.1109/DEST.2012.6227909
- [36] Nguyen, D. T., Al-Mannai, K., Joty, S. R., Sajjad, H., Imran, M. and Mitra, P., (2017b), Robust classification of crisis-related data on social networks using convolutional neural networks, ICWSM, pp. 632-635
- [37] Kabir, Md Yasin & Madria, Sanjay Kumar, (2019), A Deep Learning Approach for Tweet Classification and Rescue Scheduling for Effective Disaster Management, Proceedings of the 27th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems, (SIGSPATIAL'19), pp. 269-278, doi:10.1145/3347146.3359097
- [38] Canon, M. J., Satuito, A. & Sy, C., (2018), Determining Disaster Risk Management Priorities through a Neural Network-Based Text Classifier, 2018 International Symposium on Computer, Consumer and Control (IS3C), Taichung, Taiwan, 2018, pp. 237-241, doi:10.1109/IS3C.2018.00067
- [39] Bouzidi, Z., Amad, M. & Boudries, A. : Deep Learning-based Automated Learning Environment using Smart Data to improve Corporate Marketing, Business Strategies, Fraud Detection in Financial Services and Financial Time Series Forecasting. International Conference on Managing Business through Web Analytics - (ICMBWA2020), Khemis Miliana University, Algeria
- [40] Bouzidi, Z., Boudries, A. and Amad, M., (2021b), LSTM-based automated learning with smart data to improve marketing fraud detection and financial forecasting, Proceedings of 5th International on Economics & Management (Eman 2021) March 18, 2021, Series 49
- [41] Bouzidi, Z., Boudries, A. & Amad, Mourad, (2021e), A Hybrid of Deep CNN-LSTM Model to improve Warning, Situational Awareness and Disaster Education in Managing Emergency: Case study of Covid-19, International Journal of Business Information Systems Strategies, Inderscience Publishers, Submitted
- [42] Fiore U., et al. "Using generative adversarial networks for improving classification effectiveness in credit card fraud detection". Information Sciences 479 (2019): 448- 455.
- [43] Serpinis G., et al. "Neural networks in financial trading". Annals of Operations Research (2021): 293-308.
- [44] Calvez A (le) and Cliff D. "Deep learning can replicate adaptive traders in a limit-order-book financial market". 2018 IEEE Symposium Series on Computational Intelligence, (SSCI 2018): 1876-1883.
- [45] Kodogiannis V and Lolis A. "Forecasting Financial Time Series using Neural Network and Fuzzy System-based Techniques". Neural Computing and Applications 11 (2002): 90-102.
- [46] Fischer J., Jiang W. and Moran S., (2012), 'AtomicOrchid: A mixed reality game to investigate coordination in disaster response', In Proceedings of the 11th International Conference on Entertainment Computing (ICEC 2012), p. 572-577
- [47] Pinheiro, et al. "Stock market prediction with deep learning: A character-based neural language model for event based trading". Proceedings of the Australasian Language Technology Association Workshop (2017): 6-15.
- [48] Mohammad., et al. "Hybrid deep learning model for stock price prediction". IEEE Symposium Symposium Series on Computational Intelligence SSCI. IEEE (2018): 1837-1844.
- [49] Ryman-Tubb, N. F., Krause, P. & Garn, W. (2018). How artificial intelligence and machine learning research impacts payment card fraud detection: A survey and industry benchmark. Engineering Applications of Artificial Intelligence, 76, 130–157. <http://www.sciencedirect.com/science/article/pii/S0952197618301520>
- [50] Cavalcante, R. C., Brasileiro, R. C., Souza, V. L., Nobrega, J. P. & Oliveira, A. L. (2016). Computational intelligence and financial markets: A survey and future directions. Expert Systems with Applications, vol. 55, 194–211 Enhancing Financial Time Series Forecasting, Fraud Detection, Business . . . 25
- [51] Ali Alqahtani, Mohammed Ali, Xianghua Xie and Mark W. Jones, Deep Time-Series Clustering: A Review, Electronics 2021, 10(23), 3001; <https://doi.org/10.3390/electronics10233001>, <https://doi.org/10.3390/electronics10233001> <https://www.mdpi.com/journal/electronics>
- [52] A. Figueira and L. Oliveira, "The current state of fake news: Challenges and opportunities", Proc. Comput. Sci., vol. 121, pp. 817-825, Jan. 2017
- [53] K. Warwick, (2003), Cyborg morals cyborg values cyborg ethics, Ethics Inf. Technol., vol. 5, no. 3, pp. 131-137
- [54] S. Castillo, H. Allende-Cid, W. Palma, R. Alfaro, H. S. Ramos, C. Gonzalez, et al., (2019), Detection of bots and cyborgs in Twitter: A study on the Chilean presidential election in 2017", Proc. Int. Conf. Hum.-Comput. Interact., pp. 31
- [55] A. Balestrucci and R. D. Nicola, (2020), Credulous users and fake news: A real case study on the propagation in Twitter", Proc. IEEE Conf. Evolving Adapt. Intell. Syst. (EAIS), pp. 1-8, May 2020
- [56] M. Orabi, D. Mouheb, Z. Al Aghbari and I. Kamel, (2020), Detection of bots in social media: A systematic review, Inf. Process. Manage., vol. 57, no. 4
- [57] C.-C. Wang, (2020), Fake news and related concepts: Definitions and recent research development, Contemp. Manage. Res., vol. 16, no. 3, pp. 145-174, Sep. 202
- [58] W. Shahid, Y. Li, D. Staples, G. Amin, S. Hakak and A. Ghorbani, "Are You a Cyborg, Bot or Human?—A Survey on Detecting Fake News Spreaders," in IEEE Access, vol. 10, pp. 27069-27083, 2022, doi: 10.1109/ACCESS.2022.3157724.
- [59] Zair Bouzidi, Abdelmalek Boudries and Mourad Amad, (2021), Deep LSTM-based Model using Smart Data to Improve Awareness and Education in Marketing, Business Strategy and Financial Forecasting, MC Medical Sciences (MCMS) journal, Volume 1, Issue 5. December 2021, <https://themedicon.com/medicalsciences/volume-1-issue-5>, doi:10.55162/MCMS.2021.01.034
- [60] Young, S. D., Rivers, C. and Lewis, B., (2014), Methods of using real-time social media technologies for detection and remote monitoring of HIV outcomes, Preventive Medicine, vol. 63, pp. 112-115
- [61] Krizhevsky, A. and Sutskever, I. and Hinton, G., (2012), ImageNet classification with deep convolutional neural networks, Proceedings of Advances in Neural Information Processing Systems, vol. 25, pp. 1090-1098
- [62] LeCun, Y., Bottou, L., Bengio, Y. and Haffner, P., (1998), Gradient-based learning applied to document recognition, Proceedings of the IEEE, vol. 86, No. 11, pp. 2278-2324, doi:10.1109/5.726791
- [63] Wu, Q., Ding, K. & Huang, B., (2018), Approach for fault prognosis using recurrent neural network. Journal of Intelligent Manufacturing. 1-13. Doi:10.1007/s10845-018-1428-5
- [64] He, K., Zhang, X., Ren, S. & Sun, J., (2016b), Deep Residual Learning for Image Recognition, Proceedings of Computer Vision and Pattern Recognition (CVPR), IEEE, Las Vegas, NV, pp. 770-778, doi:10.1109/CVPR.2016.90
- [65] Zhao, J., Deng, F., Cai, Y. & Chen, J., (2018), Long short-term memory - Fully connected (LSTM-FC) neural network for PM2.5 concentration prediction, Chemosphere, vol. 220, doi:10.1016/j.chemosphere.2018.12.128
- [66] Pouyanfar, S., Tao, Y. and Tian, H., Chen, S.-C. & Shyu, M.-L., (2018), Multimodal deep learning based on multiple correspondence analysis

for disaster management, World Wide Web, vol. 22, pp. 1893-1911, doi:10.1007/s11280-018-0636-4

- [67] Haley, P. J. & Soloway, D., (1992), Extrapolation limitations of multi-layer feedforward neural networks, *Proceedings 1992 IJCNN International Joint Conference on Neural Networks*, vol. 4, Baltimore, MD, USA, pp. 25-30, doi:10.1109/IJCNN.1992.227294
- [68] Mazzia, V., Salvetti, F. & Chiaberge, M. Efficient-CapsNet: capsule network with self-attention routing. *Sci Rep* 11, 14634 (2021). <https://doi.org/10.1038/s41598-021-93977-0>
- [69] Chen, W., Sidky, H. and Ferguson, A., (2019), Capabilities and Limitations of Time-lagged Autoencoders for Slow Mode Discovery in Dynamical Systems, *The Journal of Chemical Physics*, vol. 151, No. 6, doi:10.1063/1.5112048
- [70] Wolfgang Maass, (1997), Networks of spiking neurons: The third generation of neural network models, *Neural Networks*, Volume 10, Issue 9, Pages 1659-1671, ISSN 0893-6080, [https://doi.org/10.1016/S0893-6080\(97\)00011-7](https://doi.org/10.1016/S0893-6080(97)00011-7)
- [71] D. M. J. Lazer, M. A. Baum, Y. Benkler, A. J. Berinsky, K. M. Greenhill, F. Menczer, et al., (2018), The science of fake news", *Science*, vol. 359, no. 6380, pp. 1094-1096
- [72] Creech, B., (2020), Fake news and the discursive construction of technology companies' social power. *Media, Culture & Society*, 42(6), 952-968. <https://doi.org/10.1177/0163443719899801>
- [73] J. M. Burkhardt, (2017), History of fake news, *Library Technol. Rep.*, vol. 53, no. 8, pp. 5-9, 2017.
- [74] S. B. Naeem, R. Bhatti and A. Khan, (2021), An exploration of how fake news is taking over social media and putting public health at risk, *Health Inf. Libraries J.*, vol. 38, no. 2, pp. 143-149, Jun. 2021.
- [75] X. Zhang and A. A. Ghorbani, (2020), An overview of online fake news: Characterization detection and discussion, *Inf. Process. Manage.*, vol. 57, no. 2
- [76] E. Ferrara, O. Varol, C. Davis, F. Menczer and A. Flammini, "The rise of social bots", *Commun. ACM*, vol. 59, no. 7, pp. 96-104, Jul. 2016.
- [77] M. Umer, Z. Imtiaz, S. Ullah, A. Mehmood, G. S. Choi and B.-W. On, (2020), Fake news stance detection using deep learning architecture (CNN-LSTM)", *IEEE Access*, vol. 8, pp. 156695-156706.
- [78] G. Baym and J. P. Jones, *News Parody Political Satire Across Globe*, Evanston, IL, USA:Routledge, 2013.
- [79] S. Singhal, R. R. Shah, T. Chakraborty, P. Kumaraguru and S. Satoh, "SpotFake: A multi-modal framework for fake news detection", *Proc. IEEE 5th Int. Conf. Multimedia Big Data (BigMM)*, pp. 39-47, Sep. 2019.
- [80] D. Guera and E. J. Delp, "Deepfake video detection using recurrent neural networks", *Proc. 15th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, pp. 1-6, Nov. 2018
- [81] S. Cresci, M. Petrocchi, A. Spognardi and S. Tognazzi, "Better safe than sorry: An adversarial approach to improve social bot detection", *Proc. 10th ACM Conf. Web Sci.*, pp. 47-56, Jun. 2019
- [82] R. D. Nicola, M. Petrocchi and M. Pratelli, (2021), On the efficacy of old features for the detection of new bots, *Inf. Process. Manage.*, vol. 58, no. 6, Nov. 2021
- [83] Ziani A., Azizi N., Schwab D., Zenakhra D., Aldwairi M., Chekkai N., Zemmal N. & Hadj Salah M., (2021), Deceptive Opinions Detection Using New Proposed Arabic Semantic Features. IN *Proceedia Computer Science*, Volume 189, Pages 29-36, ISSN 1877-0509, <https://doi.org/10.1016/j.procs.2021.05.067>
- [84] Berglund, M., Raiko, T., Honkala, M., Karkkainen, L., Vetek, A. & Karhunen, J., (2015), *Bidirectional Recurrent Neural Networks as Generative Models*, MIT Press, Cambridge, MA, USA
- [85] Mujeeb S. & Gupta S. (2022) Fake Account Detection in Social Media Using Big Data Analytics. In: Reddy A.B., Kiranmayee B., Mukkamala R.R., Srujan Raju K. (eds) *Proceedings of Second International Conference on Advances in Computer Engineering and Communication Systems. Algorithms for Intelligent Systems*. Springer, Singapore. https://doi.org/10.1007/978-981-16-7389-4_57
- [86] Aloysius G and Binu D., (2013), "An approach to products placement in supermarkets using Prefix Span algorithm". *Journal of King Saud University-Computer and Information Sciences* 25.1: 77-87
- [87] Han J-H., et al., (2007), "Neuronal Competition and Selection During Memory Formation". *Science* 316.5823 : 457-460
- [88] Fournier-Viger P. et al., (2017), "A Survey of Sequential Pattern Mining". *Data Science and Pattern Recognition, Ubiquitous International* 1.1: 54-78

IEEE conference templates contain guidance text for composing and formatting conference papers. Please ensure that all template text is removed from your conference paper prior to submission to the conference. Failure to remove the template text from your paper may result in your paper not being published.